

7. In a diffusion experiment, gelatin was mixed with a pink pH indicator and allowed to set. The gelatin was then cut into different sized cubes. The cubes were placed in a beaker containing dilute acid. As the acid diffused through the gelatin the cubes became colourless. The time taken to change from pink to colourless was recorded.

The surface area and volume of the cubes were calculated using the following equations:

surface area of cube = length $\times$ width $\times$ number of sides

volume of cube = length $\times$ width $\times$ height

- (a) The results of the experiment are shown below.

Length of side of gelatin cube (cm)	Time taken to become colourless (s)	Surface area of cube (cm ²)	Volume of cube (cm ³)	Surface area:volume ratio
1	20	6	1	6:1
5	41	150		6:5
7	104		343	6:7
10	610	600	1000	

- (i) **Complete** the table. [3]
- (ii) Sarah suggested that diffusion of the acid into the gelatin is quicker in the smaller cubes because the time taken for them to become colourless is less. Explain whether you agree with Sarah. [2]

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- (b) Red blood cells transport oxygen around the body to allow respiration to occur in the tissues. Red blood cells have a volume of 90 units and a surface area of 136 units. Explain how the surface area:volume ratio of red blood cells make them suitable for their function. [3]

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THE PERIODIC TABLE

Group

1 2

3

4

5

6

7

0

1 H Hydrogen 1																										
4 He Helium 2																										
		11 B Boron 5					12 C Carbon 6					14 N Nitrogen 7					16 O Oxygen 8					19 F Fluorine 9				
		27 Al Aluminium 13					28 Si Silicon 14					31 P Phosphorus 15					32 S Sulfur 16					35.5 Cl Chlorine 17				
		70 Ga Gallium 31					73 Ge Germanium 32					75 As Arsenic 33					79 Se Selenium 34					80 Br Bromine 35				
		65 Zn Zinc 30					63.5 Cu Copper 29					59 Ni Nickel 28					108 Ag Silver 47					112 Cd Cadmium 48				
		56 Fe Iron 26					101 Ru Ruthenium 44					103 Rh Rhodium 45					106 Pd Palladium 46					127 I Iodine 53				
		55 Mn Manganese 25					99 Tc Technetium 43					186 Re Rhenium 75					195 Pt Platinum 78					201 Hg Mercury 80				
		48 Ti Titanium 22					91 Zr Zirconium 40					179 Hf Hafnium 72					210 Po Polonium 84					210 At Astatine 85				
		45 Sc Scandium 21					89 Y Yttrium 39					139 La Lanthanum 57					207 Pb Lead 82					209 Bi Bismuth 83				
		39 K Potassium 19					40 Ca Calcium 20					88 Sr Strontium 38					137 Ba Barium 56					222 Rn Radon 86				
		23 Na Sodium 11					24 Mg Magnesium 12					86 Rb Rubidium 37					133 Cs Caesium 55					223 Fr Francium 87				
		7 Li Lithium 3					9 Be Beryllium 4					226 Ra Radium 88					227 Ac Actinium 89									

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number